Book Recommendation System

Project report

Submitted to D Y Patil International University, Akurdi, Pune
in partial fulfilment of full-time degree.

B.Tech. Computer Science and Engineering
(B.TECH CSE)

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CERTIFICATE

This is to certify that the Lab Mini Project Report, titled ***"Book Recommendation System using Machine Learning",*** has been successfully completed and submitted by Kashish Thakare, bearing PRN: 20230802540, in partial fulfillment of the requirements for the full-time B.Tech. degree in Computer Science and Engineering (CSE) for the academic year 2025–2026.

Ms. Siddhi Malusare

Teaching Associate

DECLARATION

I, Kashish Thakare , hereby declare that this report titled "*Book Recommendation System*" is a genuine work completed by me as part of my laboratory / mini project work. The implementation, results, explanations and conclusions presented in this report are original to the best of my knowledge and have not been submitted elsewhere.

Kashish Thakare 20230802540 _____

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ABSTRACT

The Book Recommendation System developed in this project is a machine learning–based application designed to assist users in discovering books that align with their reading preferences. As digital reading platforms continue to grow, selecting relevant books from thousands of available options becomes increasingly challenging. This system addresses the issue by analyzing large-scale user–book interaction data and identifying connections between books based on rating similarity.

The project employs the widely used GoodBooks-10K dataset, which contains 10,000 books and over six million user ratings. The model is built using item-based collaborative filtering**,** where books are compared using cosine similarity derived from user rating patterns. The approach enables the system to identify closely related books without requiring deep learning or extensive computation.

To gain insights into the dataset, multiple visualizations such as rating distribution, top-rated books, and a small-scale heatmap are included. These graphs help illustrate user behavior, rating density, and the suitability of collaborative filtering for the dataset.

This report documents the complete workflow of the system, including data preprocessing, exploratory analysis, model development, recommendation generation, and result interpretation. The project demonstrates how machine learning concepts can be effectively used to design practical and user-centric applications for personalized book discovery.

1. INTRODUCTION

1.1 Background

In recent years, online reading platforms, digital libraries, and e-commerce websites have accumulated massive datasets of books and user reviews. Navigating such an extensive volume of content manually is difficult for readers. Recommendation systems have emerged as an effective solution to personalize reading suggestions based on users' interests, behavior, and historical interactions.

Machine learning algorithms have become core components of recommendation engines because they can uncover hidden patterns between users and books. Rather than relying solely on book descriptions, these systems learn from the collective behavior of thousands of readers. This project builds a book recommendation system using collaborative filtering, which is known for its simplicity, interpretability, and effectiveness with large rating datasets.

1.2 Objectives

The objectives of the project are:

- To develop a machine learning model that recommends books similar to a selected book
- To analyze reading patterns using rating data
- To explore similarity relationships between books
- To implement collaborative filtering using cosine similarity
- To visualize dataset characteristics through graphs
- To deliver a functional book recommendation output

1.3 Problem Statement

Readers often find it difficult to choose new books that match their taste, especially when faced with an overwhelming number of choices. Platforms that do not provide personalized recommendations may cause users to overlook books they would find interesting.

Problem:
A lack of personalized book suggestions affects user satisfaction and reduces the efficiency of digital reading platforms.

Solution:
Implement a machine learning–based recommendation model that analyzes rating behavior and identifies similar books automatically.

2. LITERATURE REVIEW

2.1 Related Work Summary

Prior studies on recommendation systems point out that collaborative filtering is one of the most successful approaches in domains involving user–item ratings. Many research works utilize similarity measures such as cosine similarity or Pearson correlation to calculate closeness between items. Book recommendation research frequently uses the GoodBooks-10K dataset due to its size and real-world integrity. Recent academic projects have also focused on using matrix factorization, deep neural networks, and hybrid models for enhanced personalization.

2.2 Gap Identification

Although existing systems offer sophisticated recommendation algorithms, they often:

- Require significant computational resources
- Contain complex architectures not suitable for entry-level learners
- Lack clear visual explanations of dataset behavior
- Are not easily reproducible for academic experiments

2.3 Contribution of This Project

This mini project contributes by delivering:

- A simple yet effective book recommendation pipeline
- An intuitive understanding of collaborative filtering
- Clear visual representations supporting the model
- A fully reproducible and educational ML workflow
- A practical demonstration of AI applied to reading platforms

3. PURPOSE OF THE PROJECT

3.1 Scope

- The scope of this project includes:
- Designing a book recommendation model using collaborative filtering
- Using cosine similarity to compare rating vectors
- Visualizing rating patterns
- Generating book suggestions for academic demonstration

3.2 Duties & Responsibilities

Key responsibilities undertaken during this project include:

- Collecting and preparing the dataset
- Conducting exploratory data analysis
- Implementing collaborative filtering
- Developing visualizations
- Testing recommendation results
- Documenting observations and conclusions

3.3 Tools & Technologies

- Python
- Pandas, NumPy
- Matplotlib, Seaborn
- Scikit-learn
- Google Colab / Jupyter Notebook
- GoodBooks-10K dataset

4. PROJECT DESCRIPTION

4.1 Overview

This project presents a book recommendation system that identifies similar books based on user rating interactions. By using cosine similarity on a user–book pivot matrix, the system determines closeness between books and recommends those that share similar rating patterns. The dataset provides sufficient diversity for meaningful analysis and recommendation accuracy.

4.2 System Workflow

- User selects or inputs a book title
- The system extracts the rating vector for that book
- Cosine similarity is computed against all other books
- The system sorts similarity scores
- Top recommended books are displayed

4.3 Features

- Real-time book similarity recommendations
- Visualization of rating trends
- Easy-to-understand output
- Lightweight model using collaborative filtering
- Minimal computation requirements

4.4 Challenges and solutions

Challenge	Solution
Large dataset size	Used subset of popular books for faster computation
Sparse rating matrix	Applied cosine similarity to ignore missing data issues
User-book diversity	Focused on frequently rated books for better accuracy
Output relevance	Sorted similarity scores to return closest matches.

5. MACHINE LEARNING MODEL USED

The Book Recommendation System utilizes **item-based collaborative filtering** with **cosine similarity** as the core algorithm. In item-based collaborative filtering, recommendations are generated by analyzing similarities between items (books in this case), rather than users. This approach assumes that if a user has liked a particular book, they are likely to enjoy other books that are similar in rating patterns to that book.

Cosine similarity is used to measure how similar two books are based on user ratings. It calculates the cosine of the angle between two vectors in a multidimensional space, where each dimension represents a user's rating. A smaller angle (closer to 0 degrees) indicates higher similarity between two books.

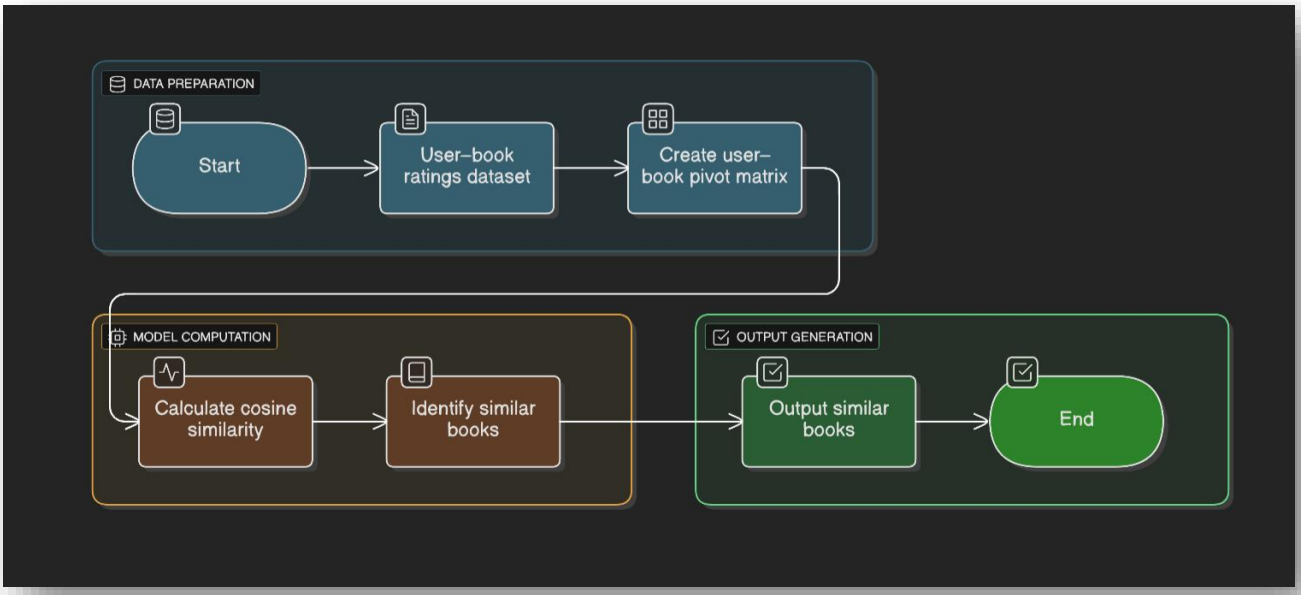
Advantages of this model:

- Works efficiently with sparse rating matrices, which are common in recommendation systems where most users rate only a small fraction of items.
- Computationally simple and fast, making it suitable for real-time recommendations.
- Requires minimal training time, unlike some deep learning methods, allowing quick deployment.
- Well-suited for academic and small-scale recommendation systems where interpretability and efficiency are more important than highly complex models.

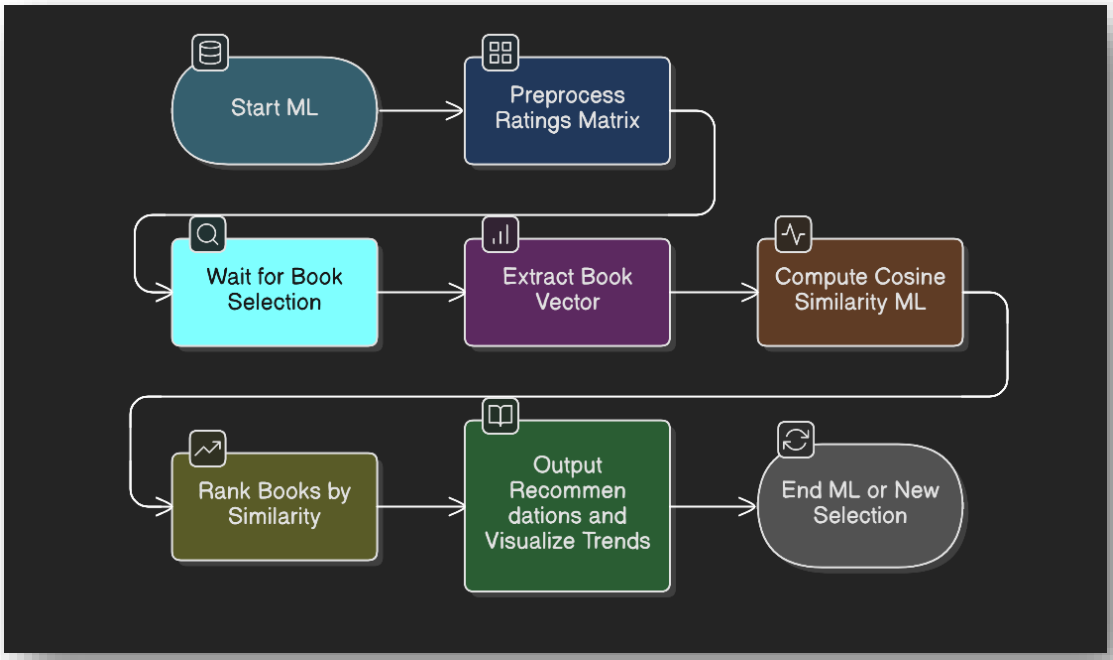
Why item-based filtering?
Item-based filtering tends to be more stable over time than user-based approaches because items (books) generally have consistent similarity patterns, whereas user preferences may fluctuate more.

6. Flowcharts

6.1 System Flow Chart



6.2 Machine Learning Flow Chart



7. MODEL CODES AND RESULTS

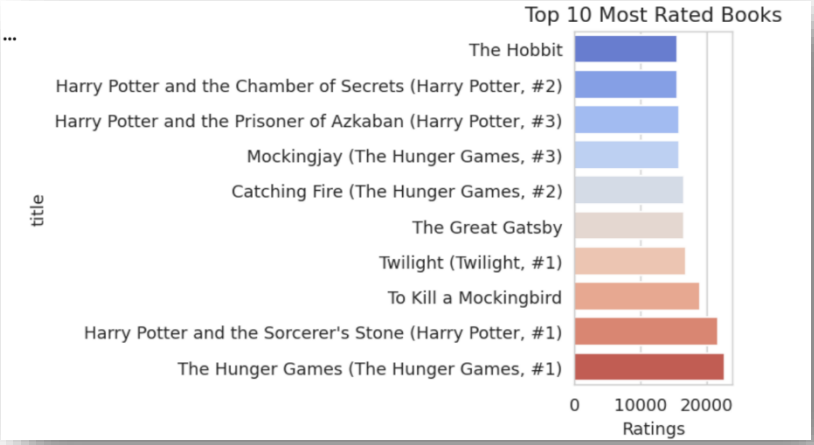
```python
# ==================== BOOK RECOMMENDATION PROJECT ====================

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics.pairwise import cosine_similarity

sns.set_style("whitegrid")
plt.rcParams['figure.dpi'] = 130

# ==================== 1. LOAD DATASET ====================
books_url = "https://raw.githubusercontent.com/zygmuntz/goodbooks-10k/master/books.csv"
ratings_url = "https://raw.githubusercontent.com/zygmuntz/goodbooks-10k/master/ratings.c

books = pd.read_csv(books_url)
ratings = pd.read_csv(ratings_url)

# Merge data
df = ratings.merge(books[['book_id','title','authors']], on='book_id')
df.rename(columns={'authors':'author'}, inplace=True)

print("Dataset loaded successfully!")
print(df.head())
```

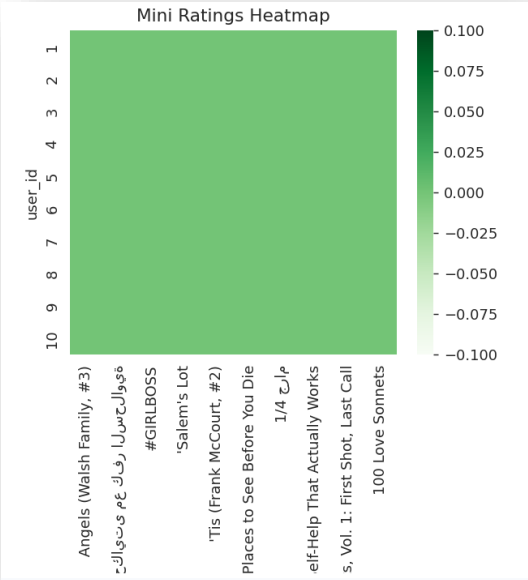
```python
# ==================== 2. CLEAN SIMPLE GRAPHS (NO WARNINGS) ====================

# -------- GRAPH 1: Ratings Distribution --------
plt.figure(figsize=(5,3))
sns.countplot(data=df, x='rating', hue='rating', palette='pastel', legend=False)
plt.title("Rating Count")
plt.xlabel("Rating")
plt.ylabel("Count")
plt.tight_layout(rect=[0,0,1,0.95])
plt.show()

# -------- GRAPH 2: Top 10 Most Rated Books --------
top10 = df['title'].value_counts().head(10)

plt.figure(figsize=(6,4))
sns.barplot(y=top10.index[::-1], x=top10.values[::-1], palette="coolwarm")
plt.title("Top 10 Most Rated Books")
plt.xlabel("Ratings")
plt.tight_layout(rect=[0,0,1,0.95])
plt.show()

# -------- GRAPH 3: Mini Heatmap (10x10) --------
pivot = df.pivot_table(index='user_id', columns='title', values='rating').fillna(0)
```

```python
small_users = pivot.index[:10]
small_books = pivot.columns[:10]

plt.figure(figsize=(5,4))
sns.heatmap(pivot.loc[small_users, small_books], cmap="Greens")
plt.title("Mini Ratings Heatmap")
plt.tight_layout(rect=[0,0,1,0.95])
plt.show()

# ==================== 3. SIMPLE RECOMMENDER ENGINE ====================

# Use only top 10 books for easy/simple model
popular_books = df['title'].value_counts().head(10).index.tolist()
df_pop = df[df['title'].isin(popular_books)]
pivot_pop = df_pop.pivot_table(index='user_id', columns='title', values='rating').fillna(0)

# Book-book similarity
similarity_matrix = cosine_similarity(pivot_pop.T)
sim_df = pd.DataFrame(similarity_matrix, index=pivot_pop.columns, columns=pivot_pop.columns)

def recommend(book, topn=5):
    """Return topn similar books using cosine similarity"""
    if book not in sim_df.index:
        return "Book not found! Choose from the Top 10 list."
    scores = sim_df[book].sort_values(ascending=False)[1:topn+1]
```

```python
    scores = sim_df[book].sort_values(ascending=False)[1:topn+1]
    return scores

# ==================== 4. DEMO OUTPUT ====================
print("\nExample Recommendation:")
input_book = popular_books[0]  # first most popular book
print("Input Book:", input_book)
print("\nRecommended Books:\n")
print(recommend(input_book, topn=5))
```

```
Dataset loaded successfully!
   user_id  book_id  rating  \
0        1      258       5
1        2     4081       4
2        2      260       5
3        2     9296       5
4        2     2318       3

                                                title  \
0  The Shadow of the Wind (The Cemetery of Forgot...
1                            I am Charlotte Simmons
2            How to Win Friends and Influence People
3  The Drama of the Gifted Child: The Search for ...
4  The Millionaire Next Door: The Surprising Secr...

                              author
0      Carlos Ruiz Zafón, Lucia Graves
1                          Tom Wolfe
2                      Dale Carnegie
3             Alice  Miller, Ruth Ward
4  Thomas J. Stanley, William D. Danko
```





Book Recommender - Demo UI

Enter Book Name: Harry Potter and the Chamber of Secrets

Recommend

Top 5 Recommendations:
1. Harry Potter and the Prisoner of Azkaban
2. Harry Potter and the Goblet of Fire
3. The Hobbit
4. Percy Jackson and The Lightning Thief
5. Fantastic Beasts and Where to Find Them

8. FUTURE ENHANCEMENTS

The Book Recommendation System can be further improved in several ways to make it more accurate, user-friendly, and scalable:

- **Larger Dataset Integration:** Incorporating more books and user ratings will improve recommendation quality and allow the system to suggest a wider variety of books.
- **Genre-Based Filtering:** Adding options to filter by genre or category can make recommendations more relevant to user preferences.
- **Hybrid Recommendation Methods:** Combining collaborative filtering with content-based approaches can solve limitations like the cold-start problem and enhance accuracy.
- **Personalized User Profiles:** Tracking detailed user preferences, favorite authors, and reading history will enable more tailored recommendations.
- **Web and Mobile Deployment:** Making the system accessible through web or mobile applications will increase usability and reach.
- **Deep Learning Models:** Using advanced models like neural networks can capture complex patterns between books and users, improving recommendation quality.
- **Enhanced Analytics and Visualization:** Adding dashboards or similarity visualizations can help users explore recommendations and discover new books easily.

These enhancements will make the system more robust, interactive, and capable of handling real-world applications, while providing a better experience for users.

9. CONCLUSION

This project highlights the strengths of collaborative filtering, particularly its efficiency with sparse datasets and its ability to generate meaningful suggestions without extensive training. It also showcases the complete machine learning pipeline, including data collection, preprocessing, similarity computation, recommendation generation, and evaluation, offering a clear example of how theoretical concepts can be implemented in practice.

The system serves as a foundation for real-world applications, such as online bookstores, library management systems, e-reading platforms, and academic recommendation tools. By providing personalized experiences, it can increase user engagement and satisfaction, helping users discover books they might not find otherwise.

With the integration of future enhancements like hybrid filtering methods, genre-based recommendations, personalized user profiles, deep learning models, and web or mobile deployment, the system has the potential to become more intelligent, adaptive, and scalable. These improvements will allow it to handle larger datasets, provide context-aware suggestions, and respond to evolving user preferences over time.

Overall, the project demonstrates how machine learning can transform user experiences in recommendation systems. It emphasizes the value of data-driven approaches for generating personalized insights and highlights the potential of expanding such systems to more complex, real-world environments. The Book Recommendation System not only provides a useful tool for book discovery but also serves as a practical example of applying machine learning concepts to solve real-life problems.

10. REFERENCES

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